

3.3 Metallic bonding homework — answers

Section A — Multiple choice

A1 = A

A2 = A

A3 = A

A4 = A

A5 = A

A6 = A

A7 = B

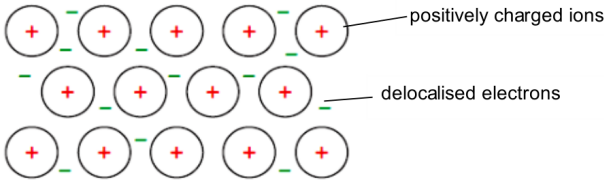
A8 = D

Section B — Mark scheme clips

B1(a) [2]

Question	Answer	Marks
2(a)	electrostatic attraction between positive metal ions / cations and delocalised electrons	2

B2(c) [1]

1(c)	 positively charged ions delocalised electrons labelled diagram shows: - regular arrangement of minimum of 2×2 circles containing + for positive ions AND surrounded by (sea of) delocalised electrons	1
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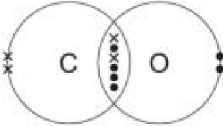
B3(a) [1]

Question	Answer	Marks
5(a)	(electrostatic) attraction between nuclei of two atoms and shared pair of electrons	1

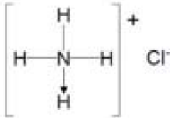
B4(b)(i) [1]

2(b)(i)	(covalent) bond with both electrons are provided from the same / one species OR shared pair (of electrons) are provided from the same species / one atom <i>owtte</i>	1
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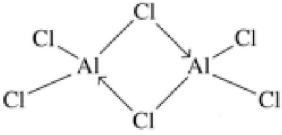
B4(b)(ii) [2]

2(b)(ii)	3 bonding pairs between C and O, 4 $\cdot$'s AND 2$\times$'s 1 lone pair on C, $\times\times$, AND 1 lone pair on O, $\bullet\bullet$. 	2
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B5(a) [2]

Question	Answer	Marks
4(a)	 OR Correct representation or label showing M1 (ionic bond between) NH_4^+ and Cl^- AND 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 M2 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons¹	2

B6(e)(i) [2]

2(e)(i)	 M1: 2 × coordinate bonds in the right place M2: all other bonds	2
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